5. The transistor has two PN junctions termed as the base-emitter junction and the base-collector junction.
6. In order to operate the transistor properly the two junctions must have the correct DC bias voltages.
7. Base-emitter junction is forward biased whereas base-collector junction is reverse biased.
8. The resistance of first junction is very small compared with the second junction. So, forward bias voltage applied to the first junction is very small compared to the second junction.

5.4.1 Transistor Lead Identification

T46KTH

There are three leads in a transistor viz. collector, emitter and base. When a transistor is to be connected in a circuit, it is necessary to know which terminal is which. The identification of the leads of transistor varies with manufacture. However, there are three systems in general use as shown in the figure 5.3.

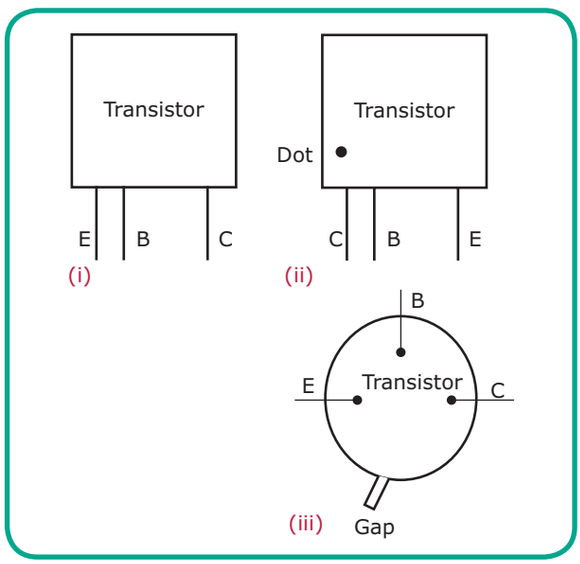


Figure 5.3 Transistor lead identification

i) When the leads of a transistor are in the same plane and unevenly spaced [see figure 5.3 (i)], they are identified by the position and spacing of leads. The centre lead is the base lead. The collector lead is identified by the large spacing existing between it and the base lead. The remaining lead is the emitter.

ii) When the leads of a transistor are in the same plane but evenly spaced [see figure 5.3 (ii)], the centre lead is the base, the lead identified by dot is the collector and the remaining lead is emitter.

iii) When the leads of a transistor are spaced around the circumference of a circle [see figure 5.3 (iii)], the three leads are generally in emitter, base, collector order clockwise from a gap.

5.5. TRANSISTOR TESTING

To test a transistor, we measure one diode junction with the multimeter leads positioned in one way and then we flip the leads of the multimeter to the reverse the polarities. One side of the diode junction should read a very high resistance of above 1MΩ of resistance (the anode to cathode side) and the other side should read a much lower resistance, may be of a few hundred thousand ohms (the cathode to anode side). This test has to be done for each junction. Figure 5.4 shows how to test a transistor.

For a single transistor there are six combination of testing the junctions as given below:

1. Emitter to Base
2. Base to Emitter
3. Emitter to Collector
4. Collector to Emitter

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